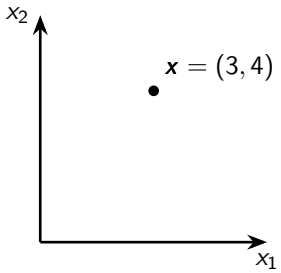
ECON 2050 – Mathematical Economics

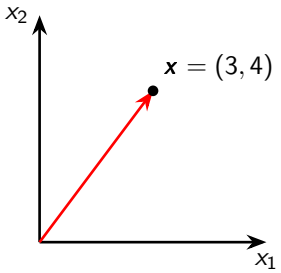
Linear Algebra I

Vectors

Loosely speaking, a vector is a term that refers to quantities that cannot be expressed by a single real number.

- More specifically, a (m-dimensional) vector is an ordered list of $m > 1$ real numbers.
- To distinguish vectors from scalars (i.e., single numbers), we denote vectors by *bold lowercase letters* (e.g., $\boldsymbol{x}$).
- A m-dimensional vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$ is an element of $\mathbb{R}^m$.





Scalar multiplication and inner product of vectors

The product of a real number (or **scalar**) $c \in \mathbb{R}$ and an (m-dimensional) vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_m) \in \mathbb{R}^m$, denoted by $c\boldsymbol{x}$, is given by

$$c\boldsymbol{x} = (cx_1, cx_2, \ldots, cx_m).$$

Also recall that in Lecture 2, we defined the **inner product** (or **dot product**) of two (m-dimensional) vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^m$ by

$$\boldsymbol{x} \cdot \boldsymbol{y} = \sum_{i=1}^{m} x_i y_i = x_1 y_1 + x_2 y_2 + \ldots + x_m y_m.$$

Linear combination

Let $S = \{\boldsymbol{x}^1, \boldsymbol{x}^2, \ldots, \boldsymbol{x}^n\}$ be a set of vectors (say in $\mathbb{R}^m$).

A vector $\boldsymbol{y}$ can be expressed as a **linear combination** of vectors in $S = \{\boldsymbol{x}^1, \boldsymbol{x}^2, \ldots, \boldsymbol{x}^n\}$ if there are real numbers $a_1, a_2, \ldots, a_n$ such that

$$\boldsymbol{y} = \sum_{j=1}^{n} a_j \, \boldsymbol{x}^j.$$

The set of all vectors that can be expressed as a linear combination of vectors in S is called the **span** of S and denoted span(S).

Example: Consider the set of vectors $S = \{(0,1),(2,1)\}$.

span(S) is the set of all linear combinations of the vectors in S. Since S contains two vectors $\boldsymbol{x}^1 = (0,1)$ and $\boldsymbol{x}^2 = (2,1)$, the span is

$$\begin{aligned}\text{span}(S) &= \{a_1(0,1) + a_2(2,1) : a_1, a_2 \in \mathbb{R}\} \\ &= \{(0 \cdot a_1 + 2 \cdot a_2, 1 \cdot a_1 + 1 \cdot a_2) : a_1, a_2 \in \mathbb{R}\} \\ &= \{(2a_2, a_1 + a_2) : a_1, a_2 \in \mathbb{R}\}\end{aligned}$$

But then any vector $\boldsymbol{x} = (x_1, x_2) \in \mathbb{R}^2$ can be written as $(2a_2, a_1 + a_2)$ by letting $a_2 = x_1/2$ and $a_1 = x_2 - x_1/2$. Thus, span$(S) = \mathbb{R}^2$.

Linear independence and dependence

The set S of vectors is said to be **linearly independent** if

$$\sum_{j=1}^{n} a_j \boldsymbol{x}^j = \mathbf{0} \quad \text{implies} \quad a_j = 0 \ \text{ for all } \ j = 1, \ldots, n,$$

where $\mathbf{0}$ denotes the zero vector $(0, 0, \ldots, 0)$. And the set S is said to be **linearly dependent** if it is not linearly independent; that is, there exist $a_1, a_2, \ldots, a_n \in \mathbb{R}$, not all zero, such that

$$\sum_{j=1}^{n} a_j \boldsymbol{x}^j = \mathbf{0}.$$

In words, a set of vectors is linearly dependent if at least one of its elements can be expressed as a linear combination of some other elements; and a set of vectors is linearly independent if none of its elements can be expressed as a linear combination of some other elements.

Example: Any set of vectors that contains only one vector that is not $\mathbf{0}$ is (trivially) linearly independent.

- For example, the set $T = \{(1, -2)\}$ is linearly independent.
- But the set $O = \{(0, 0)\}$ is linearly dependent.

Example: Consider, once again, the set of vectors $S = \{(0, 1), (2, 1)\}$. To see whether it is linearly independent, consider

$$a_1(0, 1) + a_2(2, 1) = (0, 0).$$

This can be written as $(2a_2, a_1 + a_2) = (0, 0)$, which requires $2a_2 = 0$ and $a_1 + a_2 = 0$. From the first equation, we know that $a_2 = 0$; plugging it back into the second equation, we see that $a_1 = 0$. Therefore, S must be linearly independent.

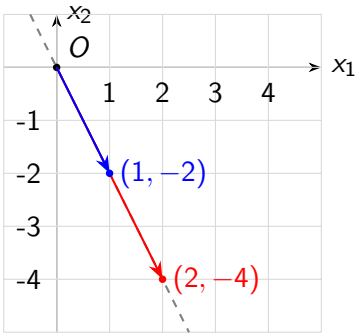
Example: Straightforwardly, the set of vectors $R = \{(1, -2), (2, -4)\}$ is linearly dependent. This is because

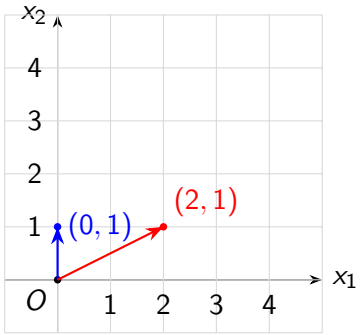
$$2(1, -2) = (2, -4) \quad \Longleftrightarrow \quad 2(1, -2) - (2, -4) = (0, 0).$$

That is, there exist $a_1 = 2$ and $a_2 = -1$ (both nonzero) such that $a_1(1, -2) + a_2(2, -4) = (0, 0)$.

Example: Consider the set of vectors $P = \{(1, 1, 0), (0, -3, 1), (2, 5, -1)\}$. This set is also linearly dependent because

$$(2, 5, -1) = 2(1, 1, 0) - (0, -3, 1) \quad \Longleftrightarrow \quad 2(1, 1, 0) - (0, -3, 1) - (2, 5, -1) = (0, 0, 0).$$





- The linearly dependent vectors $(1, -2)$ and $(2, -4)$ are "collinear" in the sense that they lie on the same line through the origin.
- The linearly independent vectors $(0, 1)$ and $(2, 1)$ are not collinear, which suffices for spanning the entire plane ($\mathbb{R}^2$).

Rank

The **rank** of a set S of vectors is the maximal number of linearly independent vectors in S, denoted by $\text{rank}(S)$.

- If S is a set of vectors in $\mathbb{R}^m$, then the discussion on the previous slide indicates that $\text{rank}(S) \leq m$.

Examples:

- $T = \{(1, -2)\}$, rank $= 1$;
- $S = \{(0, 1), (2, 1)\}$, rank $= 2$;
- $B = \{(0, 1), (1, 0)\}$, rank $= 2$;
- $R = \{(1, -2), (2, -4)\}$, rank $= 1$;
- $P = \{(1, 1, 0), (0, -3, 1), (2, 5, -1)\}$, rank $= 2$.

From vectors to matrices

Consider a vector $\boldsymbol{x} = (1, 2, 3)$. It can be written either as a "row array of numbers," $\boldsymbol{x} = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$, or as a "column array of numbers:"

$$\boldsymbol{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Consequently, a set of vectors in $\mathbb{R}^m$ can be interpreted in the following way: either take each element as a row array, or take each element as a column array, and then "stack" them together. For example, if $S = \{(1, 2, 3), (4, 5, 6)\}$, then by interpreting each element as a row array and a column array, we get, respectively, matrices

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

Conversely, it will be useful later on to think of the columns and rows of a matrix as vectors.

Matrices: Basics

A rectangular array of numbers consisting of m rows and n columns is called an $m \times n$ **matrix**.

- A matrix is usually denoted by a *bold uppercase letter* (e.g., $\boldsymbol{A}$), and the number in the i-th row and j-th column is denoted a_{ij}.
- Whenever we use lowercase letters to denote the numbers appearing in the matrix, we use the corresponding bold uppercase letters to denote the matrix.
- For a matrix $\boldsymbol{A}$, the i-th row is denoted $\boldsymbol{a}_i$ and the j-th column is denoted $\boldsymbol{a}^j$.

If $\boldsymbol{A}$ is a matrix, its **transpose**, written $\boldsymbol{A}^\mathsf{T}$ is the matrix obtained by interchanging the columns of $\boldsymbol{A}$ with its rows.

- Example: If the matrix

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} \quad \text{then} \quad \boldsymbol{A}^\mathsf{T} = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}.$$

For a vector $\boldsymbol{x} \in \mathbb{R}^m$,

- when interpreted as a row array it is equivalent to a $1 \times m$ matrix; and
- when interpreted as a column array it is equivalent to a $m \times 1$ matrix.

Thus, when we wish to emphasize the interpretation we adopt, we will call $\boldsymbol{x}$ either a "$1 \times m$ vector" or a "$m \times 1$ vector."

Defining the product of a matrix and a vector will be useful:

- The product of an $m \times n$ matrix $\boldsymbol{A}$ with a $n \times 1$ vector $\boldsymbol{x} = (x_1, \ldots, x_n)$, $\boldsymbol{Ax}$, is the $m \times 1$ vector whose i th component is $\sum_{j=1}^{n} a_{ij} x_j$.
- The i th component can also be written using inner product (dot product) notation as $\boldsymbol{a}_i \cdot \boldsymbol{x}$.
- The product of an $1 \times m$ vector $\boldsymbol{y} = (y_1, \ldots, y_m)$ and a $m \times n$ matrix $\boldsymbol{A}$, $\boldsymbol{yA}$, is the $1 \times n$ vector whose j th component is $\sum_{i=1}^{m} a_{ij} y_i$.
- The j th component can also be written using inner product (dot product) notation as $\boldsymbol{y} \cdot \boldsymbol{a}^j$.

The **column rank** of a matrix is the maximal number of its linearly independent columns. Similarly, the **row rank** is the maximal number of its linearly independent rows.

Consider an $m \times n$ matrix A. It can either be represented by its rows:

$$\boldsymbol{A} = \begin{pmatrix} \boldsymbol{a}_1 \\ \boldsymbol{a}_2 \\ \vdots \\ \boldsymbol{a}_m \end{pmatrix},$$

or by its columns: $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{a}^1 & \boldsymbol{a}^2 & \cdots & \boldsymbol{a}^n \end{pmatrix}$. Then the row rank of $\boldsymbol{A}$ is the rank of the set of vectors $A_R = \{\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_m\}$, and the column rank of $\boldsymbol{A}$ is the rank of the set of vectors $A_C = \{\boldsymbol{a}^1, \boldsymbol{a}^2, \ldots, \boldsymbol{a}^n\}$.

Theorem

Let $\boldsymbol{A}$ be an $m \times n$ matrix. Then its column rank and row rank coincide.

The formal proof of this result is beyond the scope of this course, but it makes intuitive sense:

- Let's revisit the example of the set of vectors $S = \{(1, 2, 3), (4, 5, 6)\}$.
- If we interpret each vector in S as a row, we get a matrix

$$\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$

- Interpreting each vector in S as a column, we get a matrix

$$\boldsymbol{B} = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} = \boldsymbol{A}^{\mathsf{T}}.$$

- But the rank of a set of vectors shouldn't be affected by the way we interpret the vectors.

Therefore, we define the **rank** of a matrix to be its column rank (or row rank, which is the same).

- We denote the rank of a matrix $\boldsymbol{A}$ by $\text{rank}(\boldsymbol{A})$.

Our discussion on the the previous two slides together imply the following result:

Corollary

Let $\boldsymbol{A}$ be an $m \times n$ matrix. Then $\text{rank}(\boldsymbol{A}) \leq \min\{m, n\}$.

Finally, if $\boldsymbol{A}$ is a matrix we will write $\text{span}(\boldsymbol{A})$ to denote the span of the columns of $\boldsymbol{A}$.

- For example, if the columns of matrix $\boldsymbol{A}$ are vectors $\boldsymbol{a}^1, \boldsymbol{a}^2, \boldsymbol{a}^3$, and let $A_C = \{\boldsymbol{a}^1, \boldsymbol{a}^2, \boldsymbol{a}^3\}$. Then $\text{span}(\boldsymbol{A}) = \text{span}(A_C)$.

Let $\boldsymbol{A}$ be an $m \times n$ matrix.

- We can augment a $(m \times 1)$ vector $\boldsymbol{b}$ to its right, and denote the augmented matrix by $[\boldsymbol{A}|\boldsymbol{b}]$.
- We can also augment a $m \times k$ matrix C to its right; denote the augmented matrix by $[\boldsymbol{A}|\boldsymbol{C}]$.

Example: Consider

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ 2 & 1 \\ 4 & 0 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 3 \\ 5 \\ 2 \end{pmatrix}, \quad \text{and} \quad \boldsymbol{C} = \begin{pmatrix} 0 & 1 & 0 \\ -2 & 2 & 0 \\ -2 & 3 & 0 \end{pmatrix}.$$

Then

$$[\boldsymbol{A}|\boldsymbol{b}] = \begin{pmatrix} 0 & 1 & 3 \\ 2 & 1 & 5 \\ 4 & 0 & 2 \end{pmatrix}, \quad \text{and} \quad [\boldsymbol{A}|\boldsymbol{C}] = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & -2 & 2 & 0 \\ 4 & 0 & -2 & 3 & 0 \end{pmatrix}.$$

Historically, matrices are introduced in the study of solving linear equations.

For example, the system of equations

$$\begin{cases} -4x_1 + 2x_2 - 5x_3 &= 1 \\ 2x_1 - x_2 + 2.5x_3 &= 1 \end{cases}$$

can be written as $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, where

$$\boldsymbol{A} = \begin{pmatrix} -4 & 2 & -5 \\ 2 & -1 & 2.5 \end{pmatrix}, \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Feasibility

We will be interested in problems of the following kind:

> Given an $m \times n$ matrix $\boldsymbol{A}$ and $\boldsymbol{b} \in \mathbb{R}^m$, find an $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Ax} = \boldsymbol{b}$ or prove that no such $\boldsymbol{x}$ exists.

- These problems are ultimately about whether a system of linear equations has a solution.
- The problem of whether we can find an $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Ax} = \boldsymbol{b}$ can be stated as: is $\boldsymbol{b} \in \text{span}(\boldsymbol{A})$?
- Convincing another that the system has a solution (when it does) is easy: just verify.
- However, convincing another that it *doesn't* have a solution can be much more difficult: there are infinitely many possibilities.

Consider the system of equations

$$\begin{pmatrix} -4 & 2 & -5 \\ 2 & -1 & 2.5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

and suppose we wish to know if it has a solution.

For the moment suppose that it does, and denote it by $\boldsymbol{x}^*$. Then

- Adding a linear multiple of one equation to the other yields another equation that $\boldsymbol{x}^*$ must also satisfy.
- We multiply the second equation by 2 and add it to the first.
- This produces $0x_1 + 0x_2 + 0x_3 = 3$, which cannot admit a solution.

This suggests that we might be able to decide the insolvability of a system by deriving, through appropriate linear combinations of the given equations, an inconsistency.

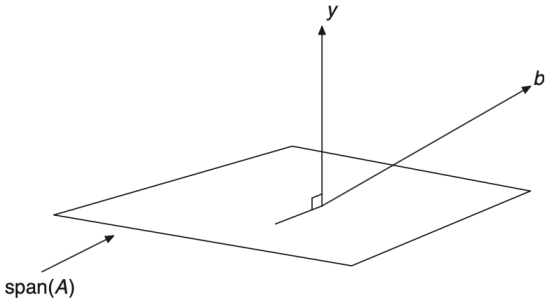
Solvability of linear equations

Theorem

Let $\boldsymbol{A}$ be an $m \times n$ matrix, $\boldsymbol{b} \in \mathbb{R}^m$, and $F = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\}$. Then either $F \neq \varnothing$ or there exists $\boldsymbol{y} \in \mathbb{R}^m$ such that $\boldsymbol{y}\boldsymbol{A} = \boldsymbol{0}$ and $\boldsymbol{y} \cdot \boldsymbol{b} \neq 0$ but not both.

In words: Consider a system of m linear equations with n unknowns. Then either the system of linear equations has a solution, or we can find m "weights," one on each of the equations, that derives an inconsistency.

Geometric interpretation:



- Suppose $F = \varnothing$. Then, $\boldsymbol{b}$ is not in the span of the columns of $\boldsymbol{A}$.
- If we think of the span of the columns of $\boldsymbol{A}$ as a plane, then $\boldsymbol{b}$ is a vector pointing out of the plane.
- Thus, any vector, $\boldsymbol{y}$, orthogonal to this plane (and so to every column of $\boldsymbol{A}$) must have a non-zero inner product (dot product) with $\boldsymbol{b}$.

Algebraic intuition:

- Take any linear combination of the equations in the system $\boldsymbol{Ax} = \boldsymbol{b}$.
- This linear combination can be obtained by multiplying each side of the system by a suitable vector $\boldsymbol{y}$: $\boldsymbol{yAx} = \boldsymbol{y} \cdot \boldsymbol{b}$.
- Any vector that solves the first system must also solve the second.
- If $\boldsymbol{yA} = \boldsymbol{0}$, then $\boldsymbol{yAx} = \boldsymbol{0} \cdot \boldsymbol{x} = 0$ for any $\boldsymbol{x} \in \mathbb{R}^n$. But then $\boldsymbol{y} \cdot \boldsymbol{b} \neq 0$ implies that $\boldsymbol{yAx} \neq \boldsymbol{y} \cdot \boldsymbol{b}$.
- Thus, there doesn't exists a vector $\boldsymbol{x}$ such that $\boldsymbol{yAx} \neq \boldsymbol{y} \cdot \boldsymbol{b}$, which in turn implies that $\boldsymbol{Ax} = \boldsymbol{b}$ is impossible.

Example: Consider, once again, the system of linear equations $\boldsymbol{Ax} = \boldsymbol{b}$, where

$$\boldsymbol{A} = \begin{pmatrix} -4 & 2 & -5 \\ 2 & -1 & 2.5 \end{pmatrix}, \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Consider a vector $\boldsymbol{y} = (y_1, y_2)$, and suppose

$$\boldsymbol{yA} = \begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} -4 & 2 & -5 \\ 2 & -1 & 2.5 \end{pmatrix} = \begin{pmatrix} -4y_1 + 2y_2 & 2y_1 - y_2 & -5y_1 + 2.5y_2 \end{pmatrix} = \boldsymbol{0};$$

this is equivalent to $2y_1 = y_2$. But $\boldsymbol{y} \cdot \boldsymbol{b} = y_1 + y_2$; if $y_1 = 1$ and $y_2 = 2$, we have $\boldsymbol{yA} = \boldsymbol{0}$ but $\boldsymbol{y} \cdot \boldsymbol{b} = 3 \neq 0$, and therefore it must be that the system of equations has no solution.

Sometimes the solvability of a system of linear equations is more conveniently determined by the following result **(will not appear in any assignments or exam)**:

Theorem

Let $\boldsymbol{A}$ be an $m \times n$ matrix of rank r and $\boldsymbol{b} \in \mathbb{R}^m$. Let r' be the rank of the augmented matrix $[\boldsymbol{A} \mid \boldsymbol{b}]$ and let $F = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\}$. Then exactly one of the following must be true:

(i) *if $r' = r + 1$, $F = \varnothing$;*

(ii) *if $r' = r = n$, F is a single vector;*

(iii) *if $r = r' < n$, then F contains infinitely many vectors.*

- $r' = r + 1$ implies that $\boldsymbol{b}$ cannot be written as a linear combination of columns of $\boldsymbol{A}$; i.e., $\boldsymbol{b} \notin \text{span}(\boldsymbol{A})$. This means that there doesn't exist $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, and hence the system has no solution.
- $r' = r$ means that $\boldsymbol{b}$ can be written as a linear combination of columns of $\boldsymbol{A}$: $\boldsymbol{b} \in \text{span}(\boldsymbol{A})$. Thus, a solution exists. If $r = n$, there are n equations and n unknowns, and therefore exactly one solution; if $r < n$, some equations are "redundant."

Example: Consider, yet another time, the system of linear equations $\boldsymbol{Ax} = \boldsymbol{b}$, where

$$\boldsymbol{A} = \begin{pmatrix} -4 & 2 & -5 \\ 2 & -1 & 2.5 \end{pmatrix}, \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Observe that

$$[\boldsymbol{A} \mid \boldsymbol{b}] = \begin{pmatrix} -4 & 2 & -5 & 1 \\ 2 & -1 & 2.5 & 1 \end{pmatrix}.$$

It can be verified that $\text{rank}(\boldsymbol{A}) = 1$, and $\text{rank}([\boldsymbol{A} \mid \boldsymbol{b}]) = 2$. Then by the theorem on the previous slide, the system of equations has no solution.

Example: Consider another system of linear equations $\boldsymbol{Ax} = \boldsymbol{c}$, where $\boldsymbol{A}$ and $\boldsymbol{x}$ are identical to the previous example, and $\boldsymbol{c} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$. Now $\text{rank}([\boldsymbol{A} \mid \boldsymbol{c}]) = 1$, and therefore the system has infinitely many solutions.